# *Phytochemicals of black bean seed coats: isolation, structure elucidation, and their antiproliferative and antioxidative activities.*

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Bioactivity-guided fractionation of black bean (Phaseolus vulgaris) seed coats was used to determine the chemical identity of bioactive constituents, which showed potent antiproliferative and antioxidative activities. Twenty-four compounds including 12 triterpenoids, 7 flavonoids, and 5 other phytochemicals were isolated using gradient solvent fractionation, silica gel and ODS columns, and semipreparative and preparative HPLC. Their chemical structures were identified using MS, NMR, and X-ray diffraction analysis. Antiproliferative activities of isolated compounds against Caco-2 human colon cancer cells, HepG2 human liver cancer cells, and MCF-7 human breast cancer cells were evaluated. Among the compounds isolated, compounds 1, 2, 6, 7, 8, 13, 14, 15, 16, 19, and 20 showed potent inhibitory activities against the proliferation of HepG2 cells, with EC50 values of 238.8 +/- 19.2, 120.6 +/- 7.3, 94.4 +/- 3.4, 98.9 +/- 3.3, 32.1 +/- 6.3, 306.4 +/- 131.3, 156.9 +/- 11.8, 410.3 +/- 17.4, 435.9 +/- 47.7, 202.3 +/- 42.9, and 779.3 +/- 37.4 microM, respectively. Compounds 1, 2, 3, 5, 6, 7, 8, 9, 10, 11, 14, 15, 19, and 20 showed potent antiproliferative activities against Caco-2 cell growth, with EC50 values of 179.9 +/- 16.9, 128.8 +/- 11.6, 197.8 +/- 4.2, 105.9 +/- 4.7, 13.9 +/- 2.8, 35.1 +/- 2.9, 31.2 +/- 0.5, 71.1 +/- 11.9, 40.8 +/- 4.1, 55.7 +/- 8.1, 299.8 +/- 17.3, 533.3 +/- 126.0, 291.2 +/- 1.0, and 717.2 +/- 104.8 microM, respectively. Compounds 5, 7, 8, 9, 11, 19, 20 showed potent antiproliferative activities against MCF-7 cell growth in a dose-dependent manner, with EC50 values of 129.4 +/- 9.0, 79.5 +/- 1.0, 140.1 +/- 31.8, 119.0 +/- 7.2, 84.6 +/- 1.7, 186.6 +/- 21.1, and 1308 +/- 69.9 microM, respectively. Six flavonoids (compounds 14-19) showed potent antioxidant activity. These results showed the phytochemical extracts of black bean seed coats have potent antioxidant and antiproliferative activities.